Technical Supplement for Compositional Automata Learning of Timed Synchronous Systems

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This document provides proofs, additional experimental analyses, implementation notes, and an illustrative learning example for the main paper. Unless stated otherwise, references to numbered sections, lemmas, and algorithms refer to the main paper.

A Proof of Lemma 1

Let $u = (d_1, \sigma_1) \cdots (d_m, \sigma_m)$. For a component $\mathcal{A}_i$, let $j_1 < \cdots < j_k$ be exactly the indices such that $\sigma_{j_\ell} \in \Sigma_i$, and let $j_0 = 0$. By the definition of projection,

$$u\downarrow_i = (D_1, \sigma_{j_1}) \cdots (D_k, \sigma_{j_k}), \qquad D_\ell = \sum_{h=j_{\ell-1}+1}^{j_\ell} d_h.$$

If there is no such index, then $u\downarrow_i = \epsilon$, which belongs to $\mathcal{L}(\mathcal{A}_i)$.

($\Rightarrow$) Assume $u \in \mathcal{L}(\mathcal{N})$ and fix a run of $\mathcal{N}$ on u. Between the actions at indices $j_{\ell-1}$ and j_ℓ, component $\mathcal{A}_i$ takes no transition. Its location remains unchanged, while the value of x_i continues to increase as time elapses. Since x_i is initially zero and reset at $j_{\ell-1}$, where $j_0 = 0$ denotes the initial instant, the value of x_i when σ_{j_ℓ} occurs, before reset, is

$$(v_{j_\ell - 1} + d_{j_\ell})(x_i) = \sum_{h=j_{\ell-1}+1}^{j_\ell} d_h = D_\ell.$$

By the NRTA run condition, the global step at j_ℓ contains a σ_{j_ℓ}-transition of $\mathcal{A}_i$ whose guard is satisfied when x_i values D_ℓ. The transitions extracted at $j_1, \ldots, j_k$ therefore form a run of $\mathcal{A}_i$ on $u\downarrow_i$. Applying the same argument to every component gives $u\downarrow_i \in \mathcal{L}(\mathcal{A}_i)$ for all $1 \le i \le n$.

($\Leftarrow$) Assume $u\downarrow_i \in \mathcal{L}(\mathcal{A}_i)$ for every component and, for each i, fix a run ρ_i of $\mathcal{A}_i$ on $u\downarrow_i$. We construct a run ρ of $\mathcal{N}$ on u by induction.

After a global prefix has been performed in ρ, the location of each component $\mathcal{A}_i$ is the location reached in ρ_i after the actions of that prefix in Σ_i. Moreover, the value of x_i equals the sum of the delays since the most recent such action, or since the initial instant if none has occurred. This correspondence holds for the empty prefix.

Suppose the correspondence holds after the first $j-1$ actions and consider (d_j, σ_j). For every i with $\sigma_j \in \Sigma_i$, the value of x_i immediately before σ_j is exactly

the next accumulated delay in $u\downarrow_i$. The next transition in ρ_i is thus an enabled σ_j-transition from its current location. All participating components can take these transitions simultaneously, while every other component keeps its location.

The update $v_j = (v_{j-1} + d_j)[\sigma_j]$ resets exactly the clocks of the participating components. It therefore preserves the correspondence after the j-th action. By induction, this construction yields a run ρ of $\mathcal{N}$ on u, and hence $u \in \mathcal{L}(\mathcal{N})$.

B Proof of Lemma 3

We say that two delays $d, d' \in \mathbb{R}_{\geq 0}$ lie in the same *RTA delay region* if, for some $k \in \mathbb{N}$, either $d = d' = k$ or $d, d' \in (k, k+1)$. Then, for any atomic constraint

$$\theta \in \{x_i \leq c,\ x_i < c,\ c \leq x_i,\ c < x_i \mid c \in \mathbb{N}\},$$

it follows that

$$\{x_i \mapsto d\} \models \theta \quad \Longleftrightarrow \quad \{x_i \mapsto d'\} \models \theta.$$

Since every guard $\phi \in \Phi(\{x_i\})$ is a conjunction of such atomic constraints, this equivalence extends to every guard in $\Phi(\{x_i\})$.

($\Rightarrow$) Let $w = (d_1, \sigma_1) \cdots (d_m, \sigma_m)$ and $w' = (d'_1, \sigma'_1) \cdots (d'_m, \sigma'_m)$, respectively. Since $[\![w]\!] = [\![w']\!]$, we have $\sigma_j = \sigma'_j$ and d_j, d'_j lie in the same RTA delay region for every $1 \leq j \leq m$. Suppose $w \in \mathcal{L}(\mathcal{A}_i)$. Then, $\mathcal{A}_i$ has a run on w using transitions $(q_{j-1}, \sigma_j, \phi_j, q_j)$ with $\{x_i \mapsto d_j\} \models \phi_j$ for every $1 \leq j \leq m$. By the guard equivalence established above, $\{x_i \mapsto d'_j\} \models \phi_j$ whenever $\{x_i \mapsto d_j\} \models \phi_j$. Hence, the same transition sequence is also a run on w', and $w' \in \mathcal{L}(\mathcal{A}_i)$.

($\Leftarrow$) The reverse implication follows by exchanging w and w' in the preceding argument, giving $w' \in \mathcal{L}(\mathcal{A}_i) \Rightarrow w \in \mathcal{L}(\mathcal{A}_i)$.

C Proof of Lemma 4

Fix a nonempty set $I \subseteq \{1, \ldots, n\}$ and a timed word $u \in (\mathbb{R}_{\geq 0} \times \Sigma_I^{loc})^*$. For every $j \notin I$, the definition of Σ_I^{loc} gives $u\downarrow_j = \epsilon$. The empty word belongs to the language of every RTA, so $u\downarrow_j \in \mathcal{L}(\mathcal{A}_j)$ and $u\downarrow_j \in \mathcal{L}(\mathcal{H}_j)$ for all $j \notin I$. By Lemma 1 applied to $\mathcal{N}$ and $\mathcal{N}_I$,

$$u \in \mathcal{L}(\mathcal{N}) \quad \Longleftrightarrow \quad u \in \mathcal{L}(\mathcal{N}_I).$$

The same argument applied to $\mathcal{N}'$ and $\mathcal{N}'_I$ gives

$$u \in \mathcal{L}(\mathcal{N}') \quad \Longleftrightarrow \quad u \in \mathcal{L}(\mathcal{N}'_I).$$

Since Algorithm 2 terminates with $\mathcal{L}(\mathcal{N}') = \mathcal{L}(\mathcal{N})$, the preceding equivalences yield

$$u \in \mathcal{L}(\mathcal{N}_I) \quad \Longleftrightarrow \quad u \in \mathcal{L}(\mathcal{N}'_I).$$

The claim follows because u was chosen arbitrarily from $(\mathbb{R}_{\geq 0} \times \Sigma_I^{loc})^*$.

D Teacher Implementation and Baseline Settings

D.1 System-Level Teacher

All three approaches use the same model-based system-level teacher, which has direct access to the target NRTA $\mathcal{N}$. For a membership query, the teacher checks whether the queried timed word belongs to $\mathcal{L}(\mathcal{N})$. For an equivalence query, it performs a breadth-first symbolic exploration of the target and hypothesis networks, representing zones by difference-bound matrices (DBMs) [1,2]. The exploration checks their language difference in both directions and reconstructs a counterexample with the fewest timed actions when a difference is found. If the exploration finds no difference, the teacher returns $\top$. Using the same backend for all approaches avoids differences caused by the teacher implementation.

D.2 Baseline Settings

Both TOT and TCT learn the target network $\mathcal{N} = (\mathcal{A}_1, \ldots, \mathcal{A}_n)$ as a single timed automaton with n clocks and require the following information:

1. the system alphabet Σ;
2. the number n of clocks;
3. the clock ceiling K_i for each clock x_i.

The first two requirements follow from $\Sigma = \bigcup_{i=1}^{n} \Sigma_i$ and the fact that every RTA component has one clock. The clock ceiling K_i is the largest integer constant in the guards of component $\mathcal{A}_i$. The region constructions in both original algorithms depend on these ceilings. We therefore supply the exact K_i values to both baselines, whereas CL neither receives nor uses them. In this paper, the timed language of an NRTA is prefix-closed, and CL exploits this property in its TOTU operations. The original TOT and TCT algorithms instead target general timed languages defined by accepting locations and do not assume prefix closure. We retain their original learning settings in our implementations.

Moreover, the *smart-teacher* variant of TOT used in our experiments and TCT require the teacher to answer *reset-information queries*, which identify the clocks reset by each action in a timed word. For an NRTA, each component resets its clock whenever it participates in an action. Hence, the reset set of an action σ is $\{x_i \mid \sigma \in \Sigma_i\}$ and can be derived directly from the known component alphabets. Therefore, reset-information queries require no additional teacher capability in our setting.

E Additional Analyses for Randomly Generated Benchmark Instances

E.1 Sensitivity to Timeout Penalties

For a timeout T and a penalty factor k, PAR-k is defined as

$$\text{PAR-}k = \frac{\sum_{c \in C_{solved}} T_c + kT\,|C_{timeout}|}{|C|},$$

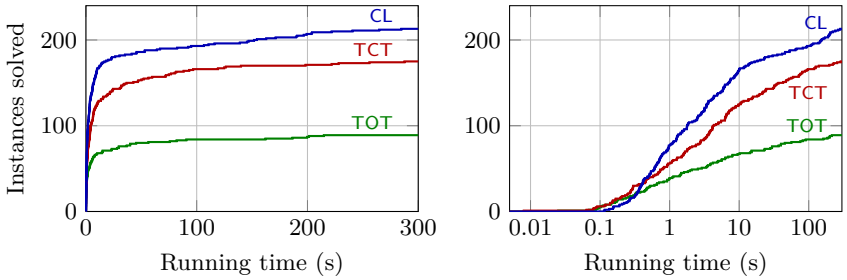
where C is the set of all benchmark instances, C_{solved} and $C_{timeout}$ are its solved and timed-out subsets, respectively, and T_c is the running time on a solved instance c. Table 3 reports the results for $T = 300$ seconds and three penalty factors. All PAR values are reported in seconds. The ordering of the approaches is unchanged for all three penalty factors, indicating that the comparison is robust to these choices.

Table 3. Sensitivity to timeout penalties.

Approach	Solved	PAR-1	PAR-2	PAR-10
TOT	89/240	196.013	384.763	1894.763
TCT	175/240	96.422	177.672	827.672
CL	213/240	54.423	88.173	358.173

E.2 Cumulative Solved Benchmark Instances

Figure 4 shows the cumulative number of randomly generated benchmark instances solved within each running-time budget. The linear scale in Fig. 4(a) shows behavior over the full time budget, while the logarithmic scale in Fig. 4(b) expands the early completion times. Both plots use observed completion times, with timed-out runs remaining unsolved.



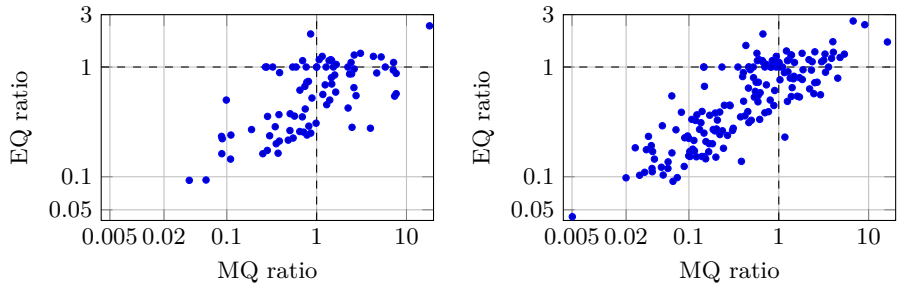
(a) Linear scale (b) Logarithmic scale

Fig. 4. Cumulative solved instances by running time.

The linear view in Fig. 4(a) shows CL continuing to solve instances over a broader range of running times and ultimately reaching a higher total, whereas the logarithmic view in Fig. 4(b) shows TOT and TCT initially accumulating solved instances faster.

E.3 Query Statistics

For brevity, we use MQ and EQ to denote system-level membership and equivalence queries, respectively. To expose the per-instance trade-off, Fig. 5 plots the ratios of the query counts of CL to those of each monolithic approach.



(a) CL/TOT (b) CL/TCT

Fig. 5. Per-instance query-count ratios.

For each jointly solved instance, the horizontal and vertical coordinates give the ratios of the MQ and EQ counts of CL to those of the corresponding monolithic approach, respectively. Thus, points farther left indicate relatively fewer MQs for CL, whereas points farther down indicate relatively fewer EQs. The dashed lines mark equal query counts. Both axes use logarithmic scales. Each point represents an instance solved by both approaches, giving 85 points in Fig. 5(a) and 162 in Fig. 5(b).

CL uses no more EQs than TOT and TCT on 70 of 85 and 129 of 162 jointly solved instances, respectively. The corresponding numbers for MQs are 47 and 114, while 45 and 103 instances satisfy both comparisons. Thus, CL's relative reduction is more consistent for EQs than MQs. This aligns with the algorithm structure, where the adapter submits an EQ only after collecting all component hypotheses, while resolving unknown TOTU entries may require additional MQs. Possible spurious counterexamples may also trigger backtracking. Overall, compositional learning changes the balance between query types rather than uniformly reducing both.

F Illustrative Example

Consider the target NRTA $\mathcal{N} = (\mathcal{A}_1, \mathcal{A}_2)$ shown in Fig. 6 with $\Sigma_1 = \{a, b\}$ and $\Sigma_2 = \{b, c\}$. The complete run uses 75 system-level membership queries and 10 system-level equivalence queries.

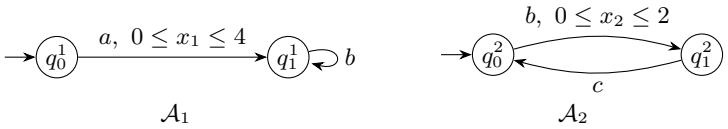


Fig. 6. Target components of the illustrative example

Let $\mathbf{T}_i^j$ denote a snapshot of the prepared TOTU of learner $\mathtt{L}_i$ at the j-th system-level equivalence query, and $\mathcal{H}_i^j$ the corresponding hypothesis. The TOTUs and component hypotheses for rounds 1–5 are shown after their corresponding descriptions. The TOTUs and hypotheses in rounds 6–10 are shown

separately. In each TOTU, the rows above the horizontal division are indexed by $\boldsymbol{S}$, and those below it are indexed by $\boldsymbol{R}$. An entry with value ? is shown as "?(β)", where $\beta \in \{\top, \bot\}$ is its concrete value in the satisfying model used to construct the hypothesis. Each system-level counterexample is shown to the right of the corresponding TOTUs, followed by its two component projections on separate lines.

Initialization. Each learner $\mathtt{L}_i$ initializes its TOTU with $\boldsymbol{S} = \boldsymbol{E} = \{\epsilon\}$ and $\boldsymbol{R} = \{([0,0], \sigma) \mid \sigma \in \Sigma_i\}$, where $f(\epsilon, \epsilon) = \top$ and all other entries have value ?. The adapter initializes each $\mathcal{P}_i$ with only its initial location and no transitions, and each backtracking stack BT_i is empty.

Round 1. The learners fill and prepare $\mathbf{T}_1^1$ and $\mathbf{T}_2^1$ and construct the corresponding hypotheses. The first equivalence query returns $u_{ce}^1 = (9/2, a) \notin \mathcal{L}(\mathcal{N})$. Its first projection has RTA region word $((4,5), a)$, while its second projection is ϵ. Since $u_{ce}^1\downarrow_1 \notin \mathcal{L}(\mathcal{P}_1)$, the adapter returns $(u_{ce}^1\downarrow_1, \bot)$ to $\mathtt{L}_1$. Since the last action a does not belong to Σ_2, it returns $(\epsilon, \top)$ to $\mathtt{L}_2$. Learner $\mathtt{L}_1$ pushes $([\![u_{ce}^1\downarrow_1]\!], \mathbf{T}_1^1)$ onto BT_1 and processes the $\bot$ feedback. Figures 7 and 8 show the corresponding TOTUs and component hypotheses.

$\mathbf{T}_1^1$	ϵ
ϵ	$\top$
$([0,0], a)$	$\top$
$([0,0], b)$	?($\top$)

$\mathbf{T}_2^1$	ϵ
ϵ	$\top$
$([0,0], c)$	$\bot$
$([0,0], b)$	?($\bot$)

$u_{ce}^1 = (9/2, a) \notin \mathcal{L}(\mathcal{N})$
$[\![u_{ce}^1\downarrow_1]\!] = ((4,5), a)$
$[\![u_{ce}^1\downarrow_2]\!] = \epsilon$

Fig. 7. TOTUs and counterexample in round 1

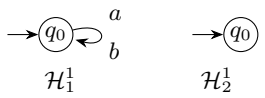


Fig. 8. Component hypotheses in round 1

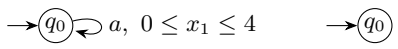
Round 2. The negative counterexample $u^2_{ce} = (0,a)(0,a)$ concerns only the first component. Since $u^2_{ce}\downarrow_1 \notin \mathcal{L}(\mathcal{P}_1)$, the adapter returns $(u^2_{ce}\downarrow_1, \bot)$ to $\mathtt{L}_1$. Since the last action a does not belong to Σ_2, it returns $(\epsilon, \top)$ to $\mathtt{L}_2$. Learner $\mathtt{L}_1$ pushes $([\![u^2_{ce}\downarrow_1]\!], \mathbf{T}^2_1)$ onto BT_1 before processing the $\bot$ feedback. Indeed, the first a moves $\mathcal{A}_1$ from q^1_0 to q^1_1, where the second a cannot occur. Figures 9 and 10 show the corresponding TOTUs and component hypotheses.

$\mathbf{T}^2_1$	ϵ
ϵ	$\top$
$((4,5),a)$	$\bot$
$([0,0],a)$	$\top$
$([0,0],b)$	$?(\bot)$

$\mathbf{T}^2_2$	ϵ
ϵ	$\top$
$([0,0],c)$	$\bot$
$([0,0],b)$	$?(\bot)$

$u^2_{ce} = (0,a)(0,a) \notin \mathcal{L}(\mathcal{N})$
$[\![u^2_{ce}\downarrow_1]\!] = ([0,0],a)([0,0],a)$
$[\![u^2_{ce}\downarrow_2]\!] = \epsilon$

Fig. 9. TOTUs and counterexample in round 2



$\mathcal{H}^2_1$ $\mathcal{H}^2_2$

Fig. 10. Component hypotheses in round 2

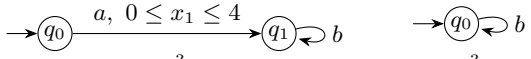
Round 3. For $u^3_{ce} = (1/2,a)(2,b) \notin \mathcal{L}(\mathcal{N})$, the last action b belongs to both component alphabets, and neither projection belongs to its positive-evidence RTA. The adapter therefore returns $(u^3_{ce}\downarrow_i, \bot)$ to each learner $\mathtt{L}_i$, which pushes $([\![u^3_{ce}\downarrow_i]\!], \mathbf{T}^3_i)$ onto BT_i before processing the $\bot$ feedback. In the target system, $u^3_{ce}\downarrow_1 \in \mathcal{L}(\mathcal{A}_1)$, whereas the accumulated delay of the projected b is $5/2$ and hence $u^3_{ce}\downarrow_2 \notin \mathcal{L}(\mathcal{A}_2)$. Thus, the component-level counterexample returned to $\mathtt{L}_1$ is spurious, whereas that returned to $\mathtt{L}_2$ is not. Figures 11 and 12 show the corresponding TOTUs and component hypotheses.

$\mathbf{T}^3_1$	ϵ	$([0,0],a)$
ϵ	$\top$	$\top$
$((4,5),a)$	$\bot$	$\bot$
$([0,0],a)$	$\top$	$\bot$
$([0,0],b)$	$?(\bot)$	$?(\bot)$
$([0,0],a)([0,0],a)$	$\bot$	$\bot$
$([0,0],a)([0,0],b)$	$\top$	$\bot$

$\mathbf{T}^3_2$	ϵ
ϵ	$\top$
$([0,0],c)$	$\bot$
$([0,0],b)$	$\top$

$u^3_{ce} = (1/2,a)(2,b) \notin \mathcal{L}(\mathcal{N})$
$[\![u^3_{ce}\downarrow_1]\!] = ((0,1),a)([2,2],b)$
$[\![u^3_{ce}\downarrow_2]\!] = ((2,3),b)$

Fig. 11. TOTUs and counterexample in round 3



$\mathcal{H}^3_1$ $\mathcal{H}^3_2$

Fig. 12. Component hypotheses in round 3

Round 4. The teacher returns the positive counterexample $u^4_{ce} = (0, a)(2, b)$. The adapter calls addPE_1 and addPE_2 on the two projections, whose RTA region words are $([0, 0], a)([2, 2], b)$ and $([2, 2], b)$, respectively, and returns $(u^4_{ce}{\downarrow_i}, \top)$ to each learner $\mathtt{L}_i$. For each $i \in \{1, 2\}$, BT_i contains no $(\pi, \mathbf{T}^k_i)$ such that π is a prefix of the corresponding RTA region word, so both learners process the $\top$ feedback without rollback. Figures 13 and 14 show the corresponding TOTUs and component hypotheses.

$\mathbf{T}^4_1$	ϵ	$([0, 0], a)$
ϵ	$\top$	$\top$
$((4, 5), a)$	$\bot$	$\bot$
$([0, 0], a)$	$\top$	$\bot$
$((0, 1), a)$	$\top$	$\bot$
$([0, 0], b)$	$\bot$	$\bot$
$((0, 1), a)([2, 2], b)$	$\bot$	$\bot$
$([0, 0], a)([0, 0], a)$	$\bot$	$\bot$
$([0, 0], a)([0, 0], b)$	$\top$	$\bot$

$\mathbf{T}^4_2$	ϵ
ϵ	$\top$
$([0, 0], c)$	$\bot$
$((2, 3), b)$	$\bot$
$([0, 0], b)$	$\top$

$u^4_{ce} = (0, a)(2, b) \in \mathcal{L}(\mathcal{N})$
$\llbracket u^4_{ce}{\downarrow_1} \rrbracket = ([0, 0], a)([2, 2], b)$
$\llbracket u^4_{ce}{\downarrow_2} \rrbracket = ([2, 2], b)$

Fig. 13. TOTUs and counterexample in round 4

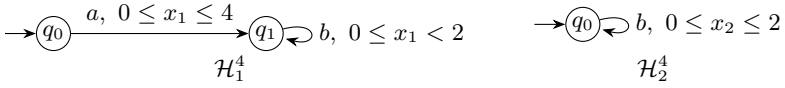


Fig. 14. Component hypotheses in round 4

Round 5. The teacher returns the negative counterexample

$$u_{ce}^5 = (0,a)(0,b)(0,b).$$

Since the last action b belongs to both component alphabets and neither projection belongs to its positive-evidence RTA, the adapter returns $(u_{ce}^5{\downarrow_i}, \bot)$ to each learner L_i. Each learner L_i pushes $([\![u_{ce}^5{\downarrow_i}]\!], \mathbf{T}_i^5)$ onto BT_i and processes the $\bot$ feedback. In fact, its first projection belongs to $\mathcal{L}(\mathcal{A}_1)$, while its second projection does not belong to $\mathcal{L}(\mathcal{A}_2)$. Figures 15 and 16 show the corresponding TOTUs and component hypotheses.

$\mathbf{T}_1^5$	ϵ	$([0,0],a)$	$([2,2],b)$
ϵ	$\top$	$\top$	$\bot$
$((0,1),a)$	$\top$	$\bot$	$\bot$
$((4,5),a)$	$\bot$	$\bot$	$\bot$
$([0,0],a)$	$\top$	$\bot$	$\top$
$([0,0],b)$	$\bot$	$\bot$	$\bot$
$((0,1),a)([0,0],a)$	$\bot$	$\bot$	$\bot$
$((0,1),a)([0,0],b)$	$\top$	$\bot$	$?(\bot)$
$((0,1),a)([2,2],b)$	$\bot$	$\bot$	$\bot$
$([0,0],a)([0,0],a)$	$\bot$	$\bot$	$\bot$
$([0,0],a)([0,0],b)$	$\top$	$\bot$	$?(\bot)$
$([0,0],a)([2,2],b)$	$\top$	$\bot$	$?(\bot)$

$\mathbf{T}_2^5$	ϵ
ϵ	$\top$
$([0,0],c)$	$\bot$
$((2,3),b)$	$\bot$
$([0,0],b)$	$\top$
$([2,2],b)$	$\top$

$u_{ce}^5 = (0,a)(0,b)(0,b) \notin \mathcal{L}(\mathcal{N})$

$[\![u_{ce}^5{\downarrow_1}]\!] = ([0,0],a)([0,0],b)([0,0],b)$

$[\![u_{ce}^5{\downarrow_2}]\!] = ([0,0],b)([0,0],b)$

Fig. 15. TOTUs and counterexample in round 5

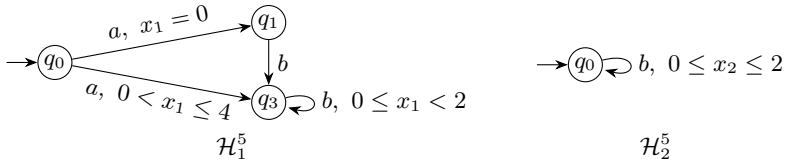


Fig. 16. Component hypotheses in round 5

Round 6. The positive counterexample

$$u^6_{ce} = (0,a)(0,b)(0,c)(0,b)$$

reveals the spurious feedback from round 5. The adapter adds both projections to the corresponding positive-evidence RTAs and returns $(u^6_{ce}\downarrow_i, \top)$ to each learner $\mathtt{L}_i$. Its first projection has RTA region word

$$\pi = ([0,0],a)([0,0],b)([0,0],b),$$

and $(\pi, \mathbf{T}^5_1)$ was pushed onto BT_1 in round 5. Learner $\mathtt{L}_1$ restores $\mathbf{T}^5_1$, pops $(\pi, \mathbf{T}^5_1)$ and all entries above it, and then processes the $\top$ feedback, which in particular sets $f(\pi, \epsilon) = \top$. The entries pushed in rounds 1–3 remain in BT_1.

For the second component,

$$[\![u^6_{ce}\downarrow_2]\!] = ([0,0],b)([0,0],c)([0,0],b).$$

BT_2 contains no $(\pi', \mathbf{T}^k_2)$ such that π' is a prefix of this RTA region word, so $\mathtt{L}_2$ processes the $\top$ feedback without rollback. For every positive projection in the remaining rounds, the corresponding backtracking stack contains no entry whose first element is its prefix, so no further rollback occurs.

$\mathbf{T}^6_1$	ϵ	$([0,0],a)$	$([2,2],b)$	$([0,0],b)$
ϵ	$\top$	$\top$	$\bot$	$\bot$
$((0,1),a)$	$\top$	$\bot$	$\bot$	$\top$
$((4,5),a)$	$\bot$	$\bot$	$\bot$	$\bot$
$([0,0],a)$	$\top$	$\bot$	$\top$	$\top$
$([0,0],a)([0,0],b)$	$\top$	$\bot$	$?(\bot)$	$\bot$
$([0,0],b)$	$\bot$	$\bot$	$\bot$	$\bot$
$((0,1),a)([0,0],a)$	$\bot$	$\bot$	$\bot$	$\bot$
$((0,1),a)([0,0],b)$	$\top$	$\bot$	$?(\bot)$	$?(\bot)$
$((0,1),a)([2,2],b)$	$\bot$	$\bot$	$\bot$	$\bot$
$([0,0],a)([0,0],a)$	$\bot$	$\bot$	$\bot$	$\bot$
$([0,0],a)([2,2],b)$	$\top$	$\bot$	$?(\bot)$	$?(\bot)$
$([0,0],a)([0,0],b)([0,0],a)$	$\bot$	$\bot$	$\bot$	$\bot$
$([0,0],a)([0,0],b)([0,0],b)$	$\bot$	$\bot$	$\bot$	$\bot$

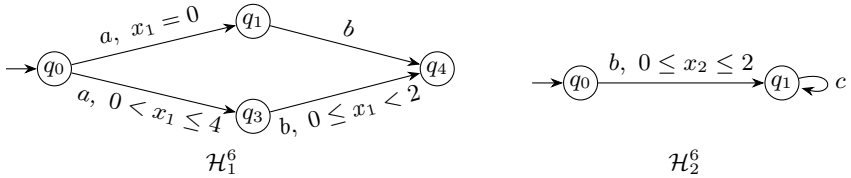
$\mathbf{T}^6_2$	ϵ	$([0,0],b)$
ϵ	$\top$	$\top$
$([0,0],b)$	$\top$	$\bot$
$([0,0],c)$	$\bot$	$\bot$
$((2,3),b)$	$\bot$	$\bot$
$([2,2],b)$	$\top$	$?(\bot)$
$([0,0],b)([0,0],b)$	$\bot$	$\bot$
$([0,0],b)([0,0],c)$	$\top$	$?(\bot)$

$u^6_{ce} = (0,a)(0,b)(0,c)(0,b) \in \mathcal{L}(\mathcal{N})$

$[\![u^6_{ce}\downarrow_1]\!] = ([0,0],a)([0,0],b)([0,0],b)$

$[\![u^6_{ce}\downarrow_2]\!] = ([0,0],b)([0,0],c)([0,0],b)$

Fig. 17. TOTUs and counterexample in round 6



$\mathcal{H}^6_1$ $\mathcal{H}^6_2$

Fig. 18. Component hypotheses in round 6

Round 7. The positive counterexample $u^7_{ce} = (0,a)(0,b)(0,c)(2,b)$ further enriches both positive-evidence RTAs. The adapter returns each projection with label $\top$ to its learner. Both learners then process the $\top$ feedback.

$\mathbf{T}^7_1$	ϵ	$([0,0],a)$	$([2,2],b)$
ϵ	$\top$	$\top$	$\bot$
$((0,1),a)$	$\top$	$\bot$	$\bot$
$((4,5),a)$	$\bot$	$\bot$	$\bot$
$([0,0],a)$	$\top$	$\bot$	$\top$
$([0,0],b)$	$\bot$	$\bot$	$\bot$
$((0,1),a)([0,0],a)$	$\bot$	$\bot$	$\bot$
$((0,1),a)([0,0],b)$	$\top$	$\bot$	$?(\bot)$
$((0,1),a)([2,2],b)$	$\bot$	$\bot$	$\bot$
$([0,0],a)([0,0],a)$	$\bot$	$\bot$	$\bot$
$([0,0],a)([0,0],b)$	$\top$	$\bot$	$?(\bot)$
$([0,0],a)([2,2],b)$	$\top$	$\bot$	$?(\bot)$
$([0,0],a)([0,0],b)([0,0],b)$	$\top$	$\bot$	$?(\bot)$

$\mathbf{T}^7_2$	ϵ	$([0,0],b)$
ϵ	$\top$	$\top$
$([0,0],b)$	$\top$	$\bot$
$([0,0],c)$	$\bot$	$\bot$
$((2,3),b)$	$\bot$	$\bot$
$([2,2],b)$	$\top$	$?(\bot)$
$([0,0],b)([0,0],b)$	$\bot$	$\bot$
$([0,0],b)([0,0],c)$	$\top$	$\top$
$([0,0],b)([0,0],c)([0,0],b)$	$\top$	$?(\bot)$

$u^7_{ce} = (0,a)(0,b)(0,c)(2,b) \in \mathcal{L}(\mathcal{N})$

$[\![u^7_{ce}\downarrow_1]\!] = ([0,0],a)([0,0],b)([2,2],b)$

$[\![u^7_{ce}\downarrow_2]\!] = ([0,0],b)([0,0],c)([2,2],b)$

Fig. 19. TOTUs and counterexample in round 7

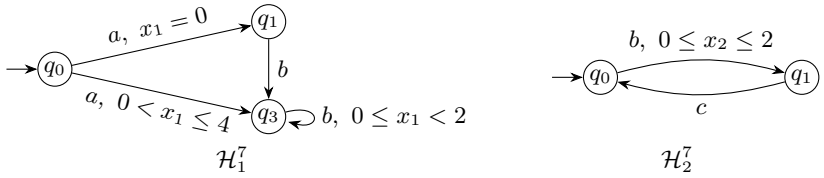


Fig. 20. Component hypotheses in round 7

Round 8. The next positive counterexample is

$$u_{ce}^8 = (0, a)(2, b)(0, c)(2, b).$$

The adapter again adds both projections to the positive-evidence RTAs and returns $(u_{ce}^8\downarrow_i, \top)$ to each learner $\mathtt{L}_i$.

$\mathbf{T}_1^8$	ϵ	$([0,0], a)$	$([2,2], b)$
ϵ	$\top$	$\top$	$\bot$
$((0,1), a)$	$\top$	$\bot$	$\bot$
$((4,5), a)$	$\bot$	$\bot$	$\bot$
$([0,0], a)$	$\top$	$\bot$	$\top$
$([0,0], b)$	$\bot$	$\bot$	$\bot$
$((0,1), a)([0,0], a)$	$\bot$	$\bot$	$\bot$
$((0,1), a)([0,0], b)$	$\top$	$\bot$	$?(\bot)$
$((0,1), a)([2,2], b)$	$\bot$	$\bot$	$\bot$
$([0,0], a)([0,0], a)$	$\bot$	$\bot$	$\bot$
$([0,0], a)([0,0], b)$	$\top$	$\bot$	$\top$
$([0,0], a)([2,2], b)$	$\top$	$\bot$	$?(\bot)$
$([0,0], a)([0,0], b)([0,0], b)$	$\top$	$\bot$	$?(\top)$
$([0,0], a)([0,0], b)([2,2], b)$	$\top$	$\bot$	$?(\bot)$

$\mathbf{T}_2^8$	ϵ	$([0,0], b)$
ϵ	$\top$	$\top$
$([0,0], b)$	$\top$	$\bot$
$([0,0], c)$	$\bot$	$\bot$
$((2,3), b)$	$\bot$	$\bot$
$([2,2], b)$	$\top$	$?(\bot)$
$([0,0], b)([0,0], b)$	$\bot$	$\bot$
$([0,0], b)([0,0], c)$	$\top$	$\top$
$([0,0], b)([0,0], c)([0,0], b)$	$\top$	$?(\bot)$
$([0,0], b)([0,0], c)([2,2], b)$	$\top$	$?(\bot)$

$u_{ce}^8 = (0, a)(2, b)(0, c)(2, b) \in \mathcal{L}(\mathcal{N})$ $\longrightarrow$

$[\![u_{ce}^8\downarrow_1]\!] = ([0,0], a)([2,2], b)([2,2], b)$

$[\![u_{ce}^8\downarrow_2]\!] = ([2,2], b)([0,0], c)([2,2], b)$

Fig. 21. TOTUs and counterexample in round 8

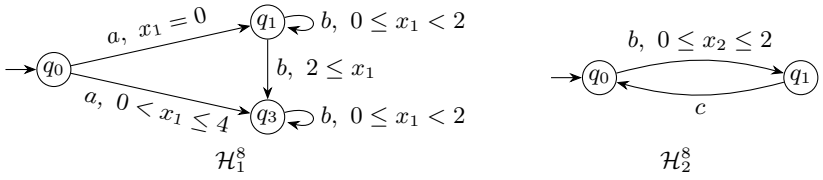


Fig. 22. Component hypotheses in round 8

Round 9. The final counterexample $u^9_{ce} = (1/3, a)(1/12, b)(1/12, c)(23/12, b)$ is positive and retains the four delays exactly. The adapter adds its projections to the positive-evidence RTAs and returns $(u^9_{ce}\downarrow_i, \top)$ to each learner $\mathtt{L}_i$.

$\mathbf{T}^9_1$	ϵ	$([0,0], a)$	$([2,2], b)$
ϵ	$\top$	$\top$	$\bot$
$((0,1), a)$	$\top$	$\bot$	$\bot$
$((4,5), a)$	$\bot$	$\bot$	$\bot$
$([0,0], a)$	$\top$	$\bot$	$\top$
$([0,0], b)$	$\bot$	$\bot$	$\bot$
$((0,1), a)([0,0], a)$	$\bot$	$\bot$	$\bot$
$((0,1), a)([0,0], b)$	$\top$	$\bot$	$?(\bot)$
$((0,1), a)([2,2], b)$	$\bot$	$\bot$	$\bot$
$([0,0], a)([0,0], a)$	$\bot$	$\bot$	$\bot$
$([0,0], a)([0,0], b)$	$\top$	$\bot$	$\top$
$([0,0], a)([2,2], b)$	$\top$	$\bot$	$\top$
$([0,0], a)([0,0], b)([0,0], b)$	$\top$	$\bot$	$?(\top)$
$([0,0], a)([0,0], b)([2,2], b)$	$\top$	$\bot$	$?(\top)$
$([0,0], a)([2,2], b)([2,2], b)$	$\top$	$\bot$	$?(\top)$

$\mathbf{T}^9_2$	ϵ	$([0,0], b)$	$([0,0], c)$
ϵ	$\top$	$\top$	$\bot$
$([0,0], b)$	$\top$	$\bot$	$\top$
$([0,0], c)$	$\bot$	$\bot$	$\bot$
$((2,3), b)$	$\bot$	$\bot$	$\bot$
$([2,2], b)$	$\top$	$?(\bot)$	$\top$
$([0,0], b)([0,0], b)$	$\bot$	$\bot$	$\bot$
$([0,0], b)([0,0], c)$	$\top$	$\top$	$\bot$
$([2,2], b)([0,0], c)$	$\top$	$?(\top)$	$\bot$
$([0,0], b)([0,0], c)([0,0], b)$	$\top$	$?(\bot)$	$\top$
$([0,0], b)([0,0], c)([2,2], b)$	$\top$	$?(\bot)$	$\top$
$([2,2], b)([0,0], c)([2,2], b)$	$\top$	$?(\bot)$	$\top$

$u^9_{ce} = (1/3, a)(1/12, b)(1/12, c)(23/12, b) \in \mathcal{L}(\mathcal{N})$

$[\![u^9_{ce}\downarrow_1]\!] = ((0,1), a)((0,1), b)([2,2], b)$

$[\![u^9_{ce}\downarrow_2]\!] = ((0,1), b)((0,1), c)((1,2), b)$

Fig. 23. TOTUs and counterexample in round 9

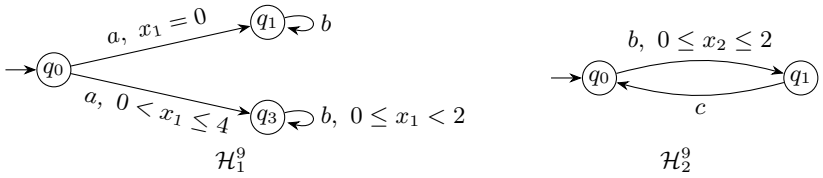


Fig. 24. Component hypotheses in round 9

Round 10. The hypotheses constructed from $\mathbf{T}_1^{10}$ and $\mathbf{T}_2^{10}$ pass the final system-level equivalence query.

$\mathbf{T}_1^{10}$	ϵ	$([0,0],a)$	$([2,2],b)$
ϵ	$\top$	$\top$	$\bot$
$((0,1),a)$	$\top$	$\bot$	$\bot$
$((4,5),a)$	$\bot$	$\bot$	$\bot$
$([0,0],a)$	$\top$	$\bot$	$\top$
$([0,0],b)$	$\bot$	$\bot$	$\bot$
$((0,1),a)([0,0],a)$	$\bot$	$\bot$	$\bot$
$((0,1),a)((0,1),b)$	$\top$	$\bot$	$\top$
$((0,1),a)([0,0],b)$	$\top$	$\bot$	$\top$
$((0,1),a)([2,2],b)$	$\bot$	$\bot$	$\bot$
$([0,0],a)([0,0],a)$	$\bot$	$\bot$	$\bot$
$([0,0],a)([0,0],b)$	$\top$	$\bot$	$\top$
$([0,0],a)([2,2],b)$	$\top$	$\bot$	$\top$
$((0,1),a)((0,1),b)([2,2],b)$	$\top$	$\bot$	$?(\top)$
$([0,0],a)([0,0],b)([0,0],b)$	$\top$	$\bot$	$?(\top)$
$([0,0],a)([0,0],b)([2,2],b)$	$\top$	$\bot$	$?(\top)$
$([0,0],a)([2,2],b)([2,2],b)$	$\top$	$\bot$	$?(\top)$

$\mathbf{T}_2^{10}$	ϵ	$([0,0],b)$	$([0,0],c)$
ϵ	$\top$	$\top$	$\bot$
$([0,0],b)$	$\top$	$\bot$	$\top$
$([0,0],c)$	$\bot$	$\bot$	$\bot$
$((0,1),b)$	$\top$	$?(\bot)$	$\top$
$((2,3),b)$	$\bot$	$\bot$	$\bot$
$([2,2],b)$	$\top$	$?(\bot)$	$\top$
$((0,1),b)((0,1),c)$	$\top$	$?(\top)$	$\bot$
$([0,0],b)([0,0],b)$	$\bot$	$\bot$	$\bot$
$([0,0],b)([0,0],c)$	$\top$	$\top$	$\bot$
$([2,2],b)([0,0],c)$	$\top$	$?(\top)$	$\bot$
$((0,1),b)((0,1),c)((1,2),b)$	$\top$	$?(\bot)$	$\top$
$([0,0],b)([0,0],c)([0,0],b)$	$\top$	$?(\bot)$	$\top$
$([0,0],b)([0,0],c)([2,2],b)$	$\top$	$?(\bot)$	$\top$
$([2,2],b)([0,0],c)([2,2],b)$	$\top$	$?(\bot)$	$\top$

$\mathsf{EQ}(\mathcal{H}_1^{10}, \mathcal{H}_2^{10}) = \top$

Fig. 25. TOTUs at the final equivalence query

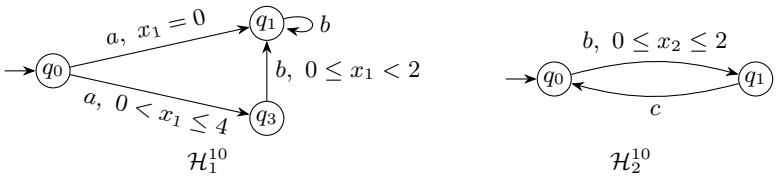


Fig. 26. Component hypotheses in round 10

The final hypotheses do not recover the target components exactly. In particular, $\mathcal{L}(\mathcal{H}_1^{10}) \subsetneq \mathcal{L}(\mathcal{A}_1)$, although the learned and target networks have the same timed language.

References

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